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Skin and Eye

Patterns of care and outcomes of proton and eye plaque brachytherapy for uveal melanoma: Review of the National Cancer Database

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ABSTRACT

PURPOSE: To examine national practice patterns and outcomes of eye plaque brachytherapy compared to proton external beam radiotherapy in the treatment of choroid melanoma.

METHODS AND MATERIALS: Demographic and clinical data for 1224 patients with choroid melanoma treated with either brachytherapy or proton beam therapy from 2004 to 2013 were obtained from the National Cancer Database. Logistic regression and propensity score matching was used to create a 1:1 matched cohort. Kaplan-Meier and Cox regression analyses were performed to evaluate survival in brachytherapy and proton groups.

RESULTS: Median followup was 37 and 29 months for brachytherapy and protons, respectively. Most patients were treated with brachytherapy (n = 996) vs. protons (n = 228). Proton patients came from more urban, affluent, and educated zip codes, and they were more likely to be treated at an academic center (all $p < 0.004$). In the propensity-score matched cohort, 2-year overall survival was 97% vs. 93%, and 5-year overall survival was 77% vs. 51% for brachytherapy and protons, respectively (p = 0.008). Multivariate Cox regression found older age (hazard ratio [HR] = 1.06, 95% confidence interval (CI) = 1.03−1.09), larger tumor diameter (12−18 mm, HR = 2.48, 95% CI = 1.40−4.42, >18 mm, HR = 6.41, 95% CI = 1.45−28.35), and protons (HR = 1.89, 95% CI = 1.06−3.37) were negative prognosticators of survival.

CONCLUSIONS: Patients selected for proton treatment have inferior survival outcomes compared to brachytherapy in this retrospective analysis. There may be unaccounted variables that influence survival, warranting further prospective studies. © 2017 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords: Uveal melanoma; Choroid melanoma; Plaque brachytherapy; Proton therapy; National Cancer Database

Introduction

Uveal melanoma accounts for 2.9% of all melanoma cases in the United States, with an age-adjusted incidence of 4.3 per million (1). It is the most common primary intraocular tumor and predominantly affects fair-skinned individuals in their 60−70s (2, 3). The Collaborative Ocular Melanoma Study (COMS) established in 2001 that overall survival (OS) was no different between patients receiving enucleation or I-125 plaque brachytherapy, with 5- and 12-year OS of 81% and 57%, respectively (4, 5).

Plaque brachytherapy and eye-preserving therapy are now the preferred treatments of uveal melanoma in the United States (3). The American Brachytherapy Society currently recommends plaque brachytherapy for tumors that are 2.5−10 mm in depth and <16 mm in diameter. Larger tumors, gross extrascleral extension, ring melanoma, or >50% involvement of the ciliary body are contraindications to plaque brachytherapy (6). Therefore, other strategies of eye-preserving therapy have been tested. A recent Surveillance, Epidemiology, and End Results Program analysis showed no difference in 5-year OS (~83%)

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